

CERTIFICATION IN

FULLSTACK DEVELOPMENT

(SPECIALIZATION IN JAVA)

DURATION: 108 HOURS

- **HTML**

- Introduction
- Elements
- Attributes
- Headings
- Link
- Text Formatting
- Style
- Image
- Table
- List
- Forms
- Iframes

- **CSS Basic**

- Introduction
- Selectors
- Color
- Background
- Fonts
- Text
- Link
- List
- Table

- **CSS Advance**

- Outline
- Cursors
- Display
- Position
- Layers
- Float

- **Bootstrap**

- Introduction
- Containers
- Grid System
- Table
- List
- Form
- Button
- Image
- Cards
- Icons
- Navbar
- Accordion
- Pagination
- Progress bar / Spinners

- **Bootstrap Advance**

- Modal
- Dropdown
- Tooltips
- Front End

- Popovers
- Alert
- Carousels
- Toasts

- **Javascript Basic**

- Variable
- Data type
- Operators
- Events
- String
- Numbers
- If ..else
- Switch Case
- Arrays
- Sorting Arrays
- Loops
- Function
- Objects

- **Javascript Advance**

- Date and Time
- Math Operations
- Type Conversions
- Events Listeners

- Borrowing Methods
- Hoisting Behavior
- Closures
- Strict Mode
- JSON Parsing
- Error Handling
- Form Validation
- Cookies
- Ajax Request

- **JQuery**

- Introduction
- Selector
- Events
- Show / Hide
- Fade
- Animation
- Stop
- Chaining
- Callback
- Get - Set
- Insert Remove
- CSS Class
- Style Properties
- Dimensions

CORE JAVA

- **Introduction to Java Programming**
 - History of Java
 - Features of Java
 - Application Design in Java
 - Simple Java Program
 - Working Environment
- **Data Types and Operators**
 - Data types
 - Naming Conventions, Variables
 - Type Casting, Operators
 - Simple Programs
- **Control Statements and Loops**
 - Need for Choices
 - If statement, If-else statement
 - Nested If-Else, Switch Case
 - Types of Loop, While Loop
 - Do -while Loop
 - For Loop

- **Arrays**

- What are Arrays?
- Types of Arrays

- **Introduction to OOPS**

- Advantages of using Array
- Identifying features of Objects
- Understanding OOPS Concept
- Advantages of using OOPS

- **Classes, Objects, Construct & Methods**

- Defining Class and Object
- Methods
- Constructors
- Use of this Keyword

- **String Class, String Buffer and String Builder**
 - Understanding String as Class
 - String Buffer
 - String Builder
 - Difference between String Buffer & String Builder
- **Encapsulation and Access Specifier**
 - Understanding Encapsulation
 - How it is Achieved
 - Access Specifiers
 - Different types of Access Specifiers
- **Inheritance**
 - Understanding Inheritance
 - How It Is Achieved
 - Types of Inheritance
 - Use of Super Keyword
 - Aggregation
- **Polymorphism**
 - Understanding Polymorphism
 - How It Is Achieved
 - Types of Polymorphism
- **Abstract Class and Interface**
 - Understanding Abstraction
 - How It Is Achieved
 - Use of Interface
 - Difference between Abstract Class & Interface

- **Exception Handling**
 - Understanding Exception
 - Types of Exception
 - Exception Handling Techniques

- **Collection Part-I**
 - Understanding Collection Framework
 - Collection Classes
 - Collection Interfaces
 - Advantages of Using Collections

- **Collection Part-II**
 - Understanding Set Collection
 - Collection Classes
 - Collection Interfaces

- **Collection Part-III**
 - Understanding Map Collection
 - Collection Classes
 - Collection Interfaces

- **File Input and Output**
 - Understanding Basic Input/Output
 - Understanding IO Stream
 - Different Types of IO Stream
 - Understanding File IO
 - Serialization

- **Multithreading**
 - Understanding Thread
 - Understanding Multithreading
 - Thread Life Cycle
 - Thread Scheduler
 - Thread Priority
- **Database Connectivity Part-I**
 - Understanding JDBC
 - Components of JDBC
 - Advantages of JDBC
- **Database Connectivity Part-II**
 - Understanding JDBC
 - Components of JDBC
 - Advantages of JDBC
- **Database Connectivity Part-III**
 - Understanding Connection
 - Understanding Statement Interface
 - Examples· ADO.NET Providers
- **Database Connectivity Part-IV**
 - Understanding Prepared Statement
 - Understanding Callable Statement
 - Transaction Management

ADVANCED JAVA

- **JSP and Servlets Overview**

- JSP and Servlets still used in 2017
- Setting Up the Development Environment Overview
- Installing Tomcat on MS Windows
- Installing Eclipse on MS Windows
- Connecting Eclipse to Tomcat
- Setting Up the Development Environment Overview
- JSP Hello World
- JSP Expressions
- JSP Fundamentals
- JSP Scriptlets
- JSP Declarations
- Calling a Java Class from JSP
- JSP Built-In Objects
- Including Files in JSP

- **Reading HTML Form Data**

- with JSP
- HTML Forms Overview - Part 1
- HTML Forms Overview - Part 2
- Drop-Down Lists
- Radio Buttons
- Checkboxes - Overview
- Checkboxes - Write some Code!
- FAQ: How to handle when user doesn't select a checkbox?

- **State Management with JSP**

- JSP
- Tracking user actions with Sessions - Overview
- ADVANCE JAVA
- Tracking user actions with Sessions - Write some Code!
- FAQ: Session Tracking - How does the TO DO LIST demo work?
- FAQ: For TO DO list example, I'm able to enter empty items. How to fix this?
- FAQ: How Do I Prevent Duplicate Items on Browser Reload?
- FAQ: Details on PageContext and Session objects
- Personalize Content with Cookies - Overview
- Personalize Content with Cookies - App Demo
- Personalize Content with Cookies - Write some Code!
- FAQ: How To Handle White-Space in Cookie Values?

- **JSP Standard Tag Library**

- JSP Tags Overview
- Install JSTL JAR files
- Where can I download the JSTL JAR files from the Internet?
- FAQ: Why Two JAR files for JSTL?
- JSTL Core Tags - Looping with forEach - Overview
- JSTL Core Tags - Project Setup
- JSTL Core Tags - Looping with forEach - Write some Code!

- JSTL Core Tags - Looping with forEach - Building HTML Tables
- FAQ: Reading Cookies with JSTL
- JSTL Core Tags - Testing Conditionals with the IF tag
- JSTL Core Tags - Testing Conditionals with the IF tag
- JSTL Core Tags - choose Tag

- **JSP Standard Tag Library (JSTL) - Function Tags**
 - JSTL Function Tags - length, toUpperCase and startsWith
 - JSTL Function Tags - split and join

- **Build a Multi-Lingual App with JSTL - i18n Tags**
 - Multi-Lingual / Internationalization Overview
 - Multi-Lingual / Internationalization Messages
 - Build a Multi-Lingual App with JSTL - Step 1
 - Build a Multi-Lingual App with JSTL - Step 2
 - Build a Multi-Lingual App with JSTL - Step 3

- **Servlet Fundamentals**
 - Hello World Servlet - Overview
 - Hello World Servlet - Let's Write some Code
 - Comparing Servlets and JSP - What's the Difference
 - Reading HTML Form Data with Servlets - Overview
 - Reading HTML Form Data with Servlets - Let's Write some Code
 - HTML Forms - Difference between GET and POST
 - Reading Servlet Parameters - Overview

- Reading Servlet Parameters - Let's Write some Code
- FAQ: Can you define per servlet parameters?
- FAQ: How To Redirect from JSP to Servlet?

- **MVC - Build an MVC App with Servlets and JSP**
 - MVC - Build an MVC App with Servlets and JSP
 - MVC with Servlets and JSP - Overview
 - MVC with Servlets and JSP - Let's Write some Code
 - MVC with Servlets and JSP - In More Detail
 - MVC with Servlets and JSP - Part 1 - Create Student Class
 - MVC with Servlets and JSP - Part 2 - Create StudentDataUtil
 - MVC with Servlets and JSP - Part 3 - Create MVC Servlet
 - MVC with Servlets and JSP - Part 4 - Create JSP View with HTML Tables

- **Build A Complete Database Web App with JDBC - Part 1**
 - Project Overview and Demo
 - Download JDBC Source Code
 - Installing the MySQL Database on MS Windows
 - Installing the MySQL Database on Mac
 - Setting up our Project Database with Sample Data
 - Setup Tomcat Datasource for Connection Pooling
 - HEADS UP - Changes for MySQL Database Server version 8

- Test Tomcat Connection Pooling
- Resources: Tune and Configure Tomcat JDBC Connection Pool for Production
- **Build A Complete Database Web App with JDBC - Part 2**
 - MVC Application Architecture
 - List Students - Overview
 - List Students - Creating the JDBC Code
 - List Students - Building the Servlet
 - List Students - Developing the JSP
 - List Students - Making it Pretty with Cascading Style Sheets (CSS)
 - List Students - Adding JSTL Functionality
 - List Students - Adding a Welcome File
- **Build A Complete Database Web App with JDBC - Part 3**
 - Add Student - Overview
 - Add Student - Setting up the Button
 - Add Student - Constructing the HTML Form
 - Add Student - Developing the Servlet
 - Add Student - Creating the JDBC Code
- **Build A Complete Database Web App with JDBC - Part 4**
 - Update Student - Overview
 - Update Student - Creating the Update Link
 - Update Student - Prepopulating the HTML Form - Part 1
 - Update Student - Prepopulating the HTML Form - Part 2
 - Update Student - Developing the Servlet

- Update Student - Prepopulating the HTML Form - Part 1
 - Update Student - Prepopulating the HTML Form - Part 2
 - Update Student - Developing the Servlet
 - Update Student - Creating the JDBC Code
- **Build A Complete Database Web App with JDBC - Part 5**
 - Delete Student - Overview
 - Delete Student - Creating the Delete Link
 - Delete Student - Developing the Servlet and JDBC Code

SQL

- **Basic Concepts**
 - Referencing Another User's Tables
 - Tables
 - Introduction of Oracle Database
 - Data Types
 - Create Database
 - Drop Database
 - Select Database
 - Create Table
 - Drop Table
 - Insert Query
 - Select Query
 - Update Query
 - Delete Query

- Order By
- Group By
- Distinct Keyword

- **Filtering, Functions, Subqueries**

- The WHERE Statement
- Filtering with AND, OR
- IN, NOT IN Statements
- Custom Columns
- Subqueries
- LIKE and MIN

- **Table Operations**

- Constraints
- Using Joins
- Unions Clause
- NULL Values
- Alias Syntax
- Indexes
- Alter Command
- Truncate Table
- Using Views
- Having Clause

- **Transactions**

- Wildcards
- Date Functions
- Temporary Tables
- Clone Tables
- Sub Queries



Happy
LEARNING!